

AWS DevOps Engineer Intern Assignment

Candidate Details

Field	Details
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Assignment Objective

Deploy a simple website on an AWS EC2 Ubuntu instance using the Nginx web server and document the complete deployment process.

Technologies Used

- Amazon EC2
- Ubuntu 24.04 LTS
- Nginx
- Linux
- HTML5
- Git
- GitHub
- Docker
- Shell Scripting

AWS Services Used

- Amazon EC2
- Security Groups

Project Architecture

Internet

Public IP

AWS Security Group
Port 22 (SSH)
Port 80 (HTTP)

Ubuntu EC2 Instance

Nginx Web Server

Static HTML Website

Task 1 – Create an AWS EC2 Instance

Steps Performed

- Logged into AWS Console.
- Opened EC2 Dashboard.
- Launched Ubuntu 24.04 LTS instance.
- Created a new Key Pair.
- Configured Security Group.
- Allowed inbound ports:
 - SSH (22)
 - HTTP (80)
- Connected to EC2 using SSH.

SSH Command

```
ssh -i "NEW.pem" ubuntu@52.66.221.35
```

Screenshots

EC2 Dashboard

Security Group

SSH Login

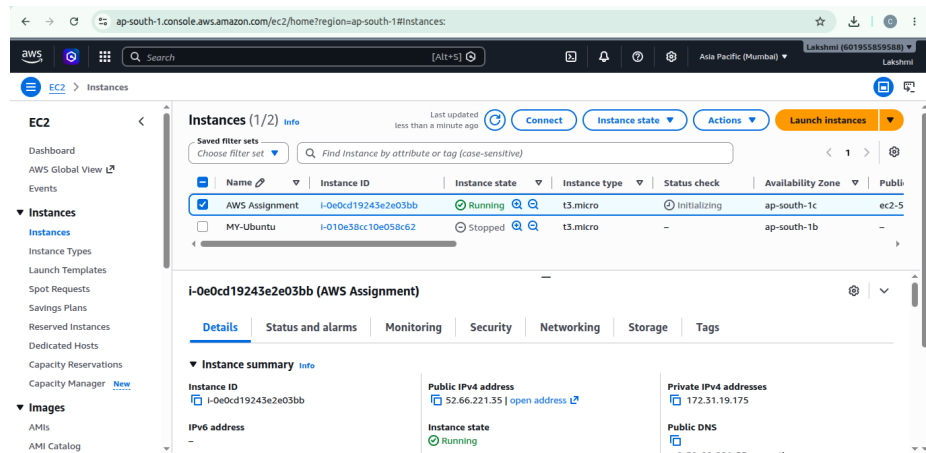


Figure 1: EC2 Dashboard

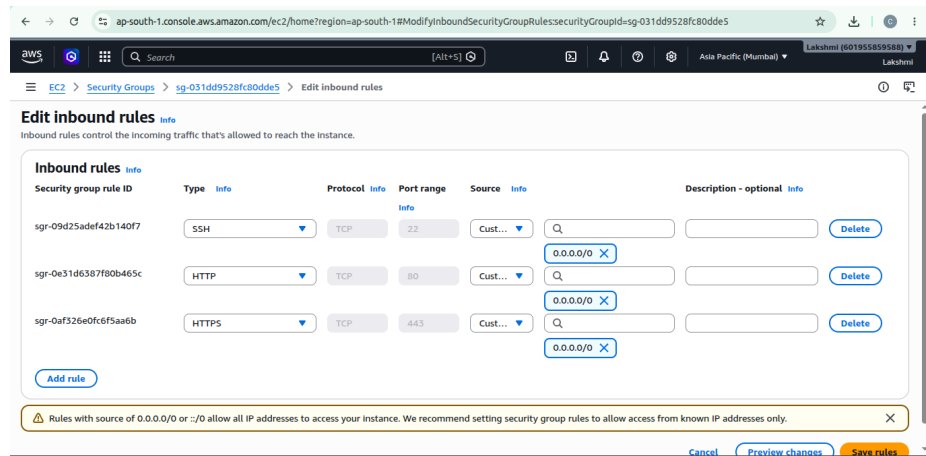


Figure 2: Security Group

```
shinbu@shinbu-Lenovo-E41-55:~/downloads$ whoami
shinbu
shinbu@shinbu-Lenovo-E41-55:~/downloads$ chmod 400 NEW.pem
shinbu@shinbu-Lenovo-E41-55:~/downloads$ ssh -i NEW.pem ubuntu@52.66.221.35
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Jul 9 16:52:40 UTC 2026

System load:  0.04      Temperature:   -273.1 C
Usage of /:   31.0% of 6.61GB    Processes:    118
Memory usage: 26%      Users logged in: 0
Swap usage:   0%          IPv4 address for ens5: 172.31.19.175

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Thu Jul 9 16:52:00 2026 from 49.37.130.49
ubuntu@ip-172-31-19-175:~$ whoami
ubuntu
ubuntu@ip-172-31-19-175:~$
```

Figure 3: SSH Login

Task 2 – Linux Basics

Update System

```
sudo apt update
sudo apt upgrade -y
```

Screenshot

```
ubuntu@ip-172-31-19-175:~$ sudo apt update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 https://download.docker.com/linux/ubuntu resolute InRelease [52.5 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [338 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [53.1 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [282 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [153 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.2 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Fetched 1120 kB in 2s (642 kB/s)
56 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-19-175:~$
```

Figure 4: System Update

Install Nginx

```
sudo apt install nginx -y
```

Screenshot

```
Setting up nginx (1.28.3-2ubuntu1.6) ...
* Upgrading binary nginx
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for ufw (0.36.2-9build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-19-175:~$ sudo status nginx
sudo: 'status': command not found
ubuntu@ip-172-31-19-175:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-09 16:59:29 UTC; 1min 25s ago
     Invocation: f7cae129a1d44a7b7e612c05fa7d0e6
       Docs: man:nginx(8)
    Process: 2280 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 2282 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 2311 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7.3M)
        CPU: 43ms
     CGroup: /system.slice/nginx.service
             └─2311 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
                 └─2314 "nginx: worker process"
                     └─2315 "nginx: worker process"

Jul 09 16:59:29 ip-172-31-19-175 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jul 09 16:59:29 ip-172-31-19-175 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
```

Figure 5: Nginx Installation

Check Nginx Status

```
sudo systemctl status nginx
```

Screenshot

```
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-19-175:~$ sudo status nginx
sudo: 'status': command not found
ubuntu@ip-172-31-19-175:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-09 16:59:29 UTC; 1min 25s ago
     Invocation: f7cae129a1d44a7b7e612c05fa7d0e6
       Docs: man:nginx(8)
    Process: 2280 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 2282 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 2311 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7.3M)
        CPU: 43ms
     CGroup: /system.slice/nginx.service
             └─2311 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
                 └─2314 "nginx: worker process"
                     └─2315 "nginx: worker process"

Jul 09 16:59:29 ip-172-31-19-175 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jul 09 16:59:29 ip-172-31-19-175 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-19-175:~$ sudo systemctl start nginx
ubuntu@ip-172-31-19-175:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-172-31-19-175:~$ nano Basic.html
ubuntu@ip-172-31-19-175:~$ ls
Basic.html
```

Figure 6: Nginx Status

Restart Nginx

```
sudo systemctl restart nginx
```

Check Disk Usage

`df -h`

Screenshot

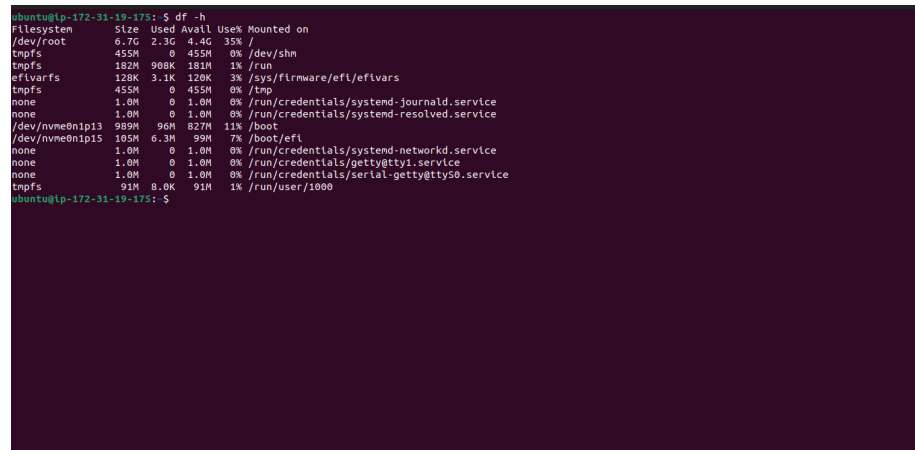


Figure 7: Disk Usage

Check Memory Usage

`free -h`

Screenshot

Running Processes

`ps aux`

or

`top`

Screenshot

```

ubuntu@ip-172-31-19-175:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.3G  4.4G  35% /
tmpfs            455M   0  455M   0% /dev/shm
tmpfs           182M  908K  181M   1% /run
efivarfs        128K   3.1K  128K   3% /sys/firmware/efi/efivars
tmpfs           455M   0  455M   0% /tmp
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13 989M  96M  827M  11% /boot
/dev/nvme0n1p15 105M   6.3M   99M   7% /boot/efi
none            1.0M   0  1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
none            1.0M   0  1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs           91M   0.0K   91M   1% /run/user/1000

ubuntu@ip-172-31-19-175:~$ free -h
             total        used        free      shared  buff/cache   available
Mem:          908Mi       335Mi       174Mi       2.7Mi       513Mi       573Mi
Swap:          0B           0B

ubuntu@ip-172-31-19-175:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  1.7 25544 16236 ?        Ss   12:00   0:04 /sbin/init
root         2  0.0  0.0   0     0 ?        S    12:00   0:00 [kthreadd]
root         3  0.0  0.0   0     0 ?        S    12:00   0:00 [pool_workqueue_release]
root         4  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-rcu_gp]
root         5  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-sync_wq]
root         6  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-kvfree_rcu_reclaim]
root         7  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-slub_flushq]
root         8  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-netns]
root        10  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/0:0H-kblockd]
root        13  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-mm_percpu_wq]
root        14  0.0  0.0   0     0 ?        S    12:00   0:00 [ksoftirqd/0]
root        15  0.0  0.0   0     0 ?        I    12:00   0:00 [rcu_sched]
root        16  0.0  0.0   0     0 ?        S    12:00   0:00 [rcu_exp_par_gp_kthread_worker/0]
root        17  0.0  0.0   0     0 ?        S    12:00   0:00 [rcu_exp_gp_kthread_worker]
root        18  0.0  0.0   0     0 ?        S    12:00   0:00 [migration/0]
root        19  0.0  0.0   0     0 ?        S    12:00   0:00 [kprobes-optimizac]

```

Figure 8: Memory Usage

```

ubuntu@ip-172-31-19-175:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.3G  4.4G  35% /
tmpfs            455M   0  455M   0% /dev/shm
tmpfs           182M  908K  181M   1% /run
efivarfs        128K   3.1K  128K   3% /sys/firmware/efi/efivars
tmpfs           455M   0  455M   0% /tmp
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13 989M  96M  827M  11% /boot
/dev/nvme0n1p15 105M   6.3M   99M   7% /boot/efi
none            1.0M   0  1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
none            1.0M   0  1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs           91M   0.0K   91M   1% /run/user/1000

ubuntu@ip-172-31-19-175:~$ free -h
             total        used        free      shared  buff/cache   available
Mem:          908Mi       335Mi       174Mi       2.7Mi       513Mi       573Mi
Swap:          0B           0B

ubuntu@ip-172-31-19-175:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  1.7 25544 16236 ?        Ss   12:00   0:04 /sbin/init
root         2  0.0  0.0   0     0 ?        S    12:00   0:00 [kthreadd]
root         3  0.0  0.0   0     0 ?        S    12:00   0:00 [pool_workqueue_release]
root         4  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-rcu_gp]
root         5  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-sync_wq]
root         6  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-kvfree_rcu_reclaim]
root         7  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-slub_flushq]
root         8  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-netns]
root        10  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/0:0H-kblockd]
root        13  0.0  0.0   0     0 ?        I<   12:00   0:00 [kworker/R-mm_percpu_wq]
root        14  0.0  0.0   0     0 ?        S    12:00   0:00 [ksoftirqd/0]
root        15  0.0  0.0   0     0 ?        I    12:00   0:00 [rcu_sched]
root        16  0.0  0.0   0     0 ?        S    12:00   0:00 [rcu_exp_par_gp_kthread_worker/0]
root        17  0.0  0.0   0     0 ?        S    12:00   0:00 [rcu_exp_gp_kthread_worker]
root        18  0.0  0.0   0     0 ?        S    12:00   0:00 [migration/0]
root        19  0.0  0.0   0     0 ?        S    12:00   0:00 [kprobes-optimizac]

```

Figure 9: Running Processes

Task 3 – Host a Simple Website

Created a basic HTML webpage containing:

- Name
- College
- Branch
- Email
- Current Date

HTML File Location

/var/www/html/Basic.html

Edit HTML

```
sudo nano /var/www/html/Basic.html
```

Restart Nginx

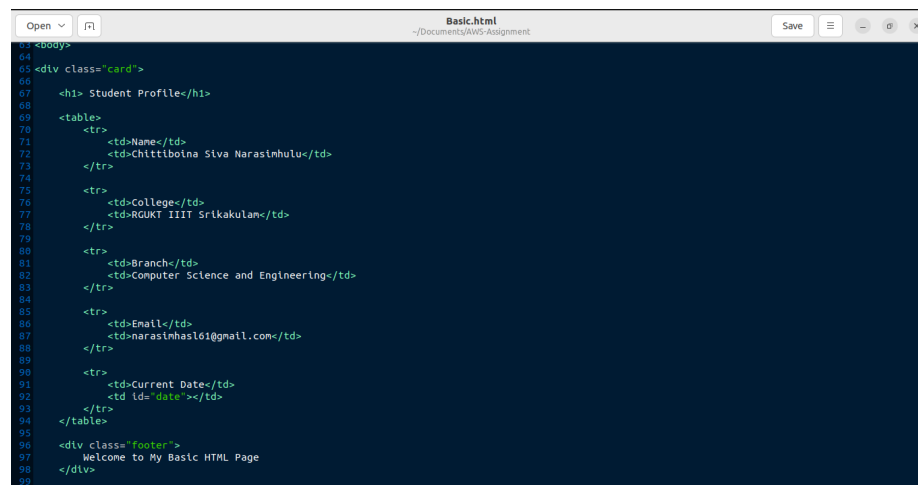
```
sudo systemctl restart nginx
```

Website URL

http://13.202.245.239/Basic.html

Screenshots

HTML File



```
64 <body>
65 <div class="card">
66   <h1> Student Profile</h1>
67   <table>
68     <tr>
69       <td>Name</td>
70       <td>Chittiboina Siva Narasinhulu</td>
71     </tr>
72     <tr>
73       <td>College</td>
74       <td>RGUKI IIII SrIkakulam</td>
75     </tr>
76     <tr>
77       <td>Branch</td>
78       <td>Computer Science and Engineering</td>
79     </tr>
80     <tr>
81       <td>Email</td>
82       <td>narasinhasi6@gmail.com</td>
83     </tr>
84     <tr>
85       <td>Current Date</td>
86       <td id="date"></td>
87     </tr>
88   </table>
89   <div class="footer">
90     Welcome to My Basic HTML Page
91   </div>
```

Figure 10: HTML File

Website Output

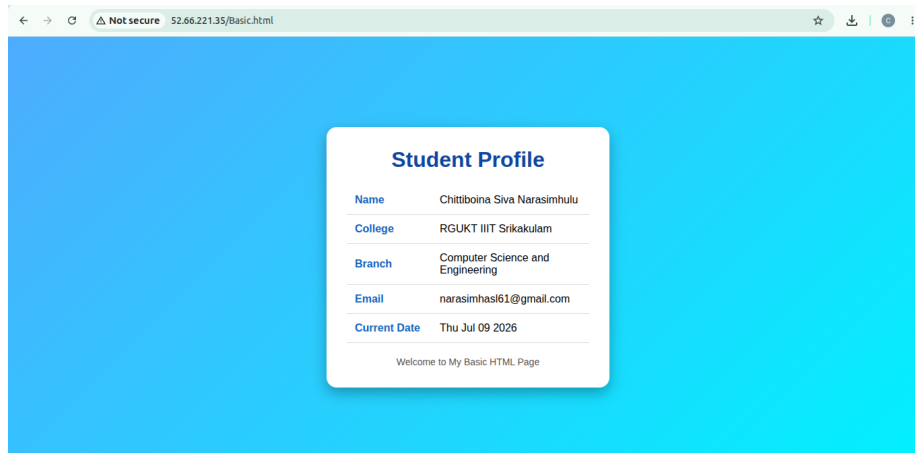


Figure 11: Website Output

Task 4 – Git & GitHub

Initialize Repository

```
git init
```

Add Files

```
git add .
```

Commit

```
git commit -m "Initial Commit"
```

Add Remote

```
git remote add origin https://github.com/sivanarasimhulu21/AWS-Assignment.git
```

Push Repository

```
git push -u origin main
```

Repository Structure

AWS-Assignment/

README.md
Basic.html
shell.sh

screenshots/

01-ec2-dashboard.png
02-security-group.png
03-ssh-login.png
04-system-update.png
05-nginx-install.png
06-nginx-status.png
07-disk-usage.png
08-memory-usage.png
09-processes.png
10-html-page.png
11-browser-output.png
12-docker-install.png
13-docker-hello-world.png
14-shell-script.png
15-shell-script-output.png

Task 5 – Documentation

The PDF report includes:

- AWS Services Used
 - EC2 Instance Setup
 - Security Groups
 - Linux Commands Used
 - Nginx Installation
 - Website Deployment
 - GitHub Repository
 - Problems Faced
 - Learnings
 - Time Taken
-

Linux Commands Used

Update Packages

```
sudo apt update  
sudo apt upgrade -y
```

Install Nginx

```
sudo apt install nginx -y
```

Start Nginx

```
sudo systemctl start nginx
```

Check Nginx Status

```
sudo systemctl status nginx
```

Restart Nginx

```
sudo systemctl restart nginx
```

Disk Usage

```
df -h
```

Memory Usage

```
free -h
```

Running Processes

```
ps aux
```

Edit Website

```
sudo nano /var/www/html/Basic.html
```

Bonus Task 1 – Install Docker

Install Docker

```
sudo apt update  
sudo apt install docker.io -y
```

Verify Docker

```
docker --version
```

Run Hello World Container

```
sudo docker run hello-world
```

Screenshots

Docker Installation

```
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-19-175:~$ sudo systemctl start docker
ubuntu@ip-172-31-19-175:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-09 17:22:32 UTC; 28s ago
     Invocation: b77d7535f0c04210b1de7647d7d4413
   TriggeredBy: ● docker.socket
      Docs: https://docs.docker.com
     Main PID: 3547 (dockerd)
        Tasks: 9
       Memory: 27.6M (peak: 28M)
          CPU: 338ms
       CGroup: /system.slice/docker.service
               └─3547 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Jul 09 17:22:32 ip-172-31-19-175 dockerd[3547]: time="2026-07-09T17:22:32.349465319Z" level=info msg="Restoring containers: start."
Jul 09 17:22:32 ip-172-31-19-175 dockerd[3547]: time="2026-07-09T17:22:32.389464249Z" level=info msg="Deleting nftables IPv4 rules" error="exit statu
Jul 09 17:22:32 ip-172-31-19-175 dockerd[3547]: time="2026-07-09T17:22:32.394732950Z" level=info msg="Deleting nftables IPv6 rules" error="exit statu
Jul 09 17:22:32 ip-172-31-19-175 dockerd[3547]: time="2026-07-09T17:22:32.693346708Z" level=info msg="Loading containers: done."
Jul 09 17:22:32 ip-172-31-19-175 dockerd[3547]: time="2026-07-09T17:22:32.709691463Z" level=info msg="Docker daemon" commit=29.1.3-0ubuntu4.1 contain
Jul 09 17:22:32 ip-172-31-19-175 dockerd[3547]: time="2026-07-09T17:22:32.709813359Z" level=info msg="Initializing buildkit"
Jul 09 17:22:32 ip-172-31-19-175 dockerd[3547]: time="2026-07-09T17:22:32.753806562Z" level=info msg="Completed buildkit initialization"
Jul 09 17:22:32 ip-172-31-19-175 dockerd[3547]: time="2026-07-09T17:22:32.762822619Z" level=info msg="Daemon has completed initialization"
Jul 09 17:22:32 ip-172-31-19-175 dockerd[3547]: time="2026-07-09T17:22:32.763405962Z" level=info msg="API listen on /run/docker.sock"
Jul 09 17:22:32 ip-172-31-19-175 systemd[1]: Started docker.service - Docker Application Container Engine.
ubuntu@ip-172-31-19-175:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-19-175:~$ docker ps
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
ubuntu@ip-172-31-19-175:~$ exit
logout
Connection to ip-172-31-19-175 closed.
```

Figure 12: Docker Installation

Docker Hello World

Bonus Task 2 – Shell Script

shell.sh

```
#!/bin/bash

# Restart Nginx service

echo "Restarting Nginx..."

sudo systemctl restart nginx

if [ $? -eq 0 ]; then
```

```
ubuntu@ip-172-31-19-175:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
ubuntu@ip-172-31-19-175:~$ docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
4f550867fdd0: Pull complete
d5e71ed42bfs: Download complete
Digest: sha256:96498fffd522e70807ab6384asc0485a79b9c7c08ca79ba08623edcad1054e62d
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
ubuntu@ip-172-31-19-175:~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (and64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
ubuntu@ip-172-31-19-175:~$
```

Figure 13: Docker Hello World

```
echo "Nginx restarted successfully."
else
echo "Failed to restart Nginx."
fi
```

Make Executable

```
chmod +x shell.sh
```

Execute Script

```
./shell.sh
```

Screenshots

Shell Script Creation

Shell Script Output

Problems Faced

- SSH permission denied while connecting to EC2.
 - GitHub authentication issue while pushing repository.
 - Replacing the default Nginx page.
 - File permission issues while editing `/var/www/html/index.html`.
-

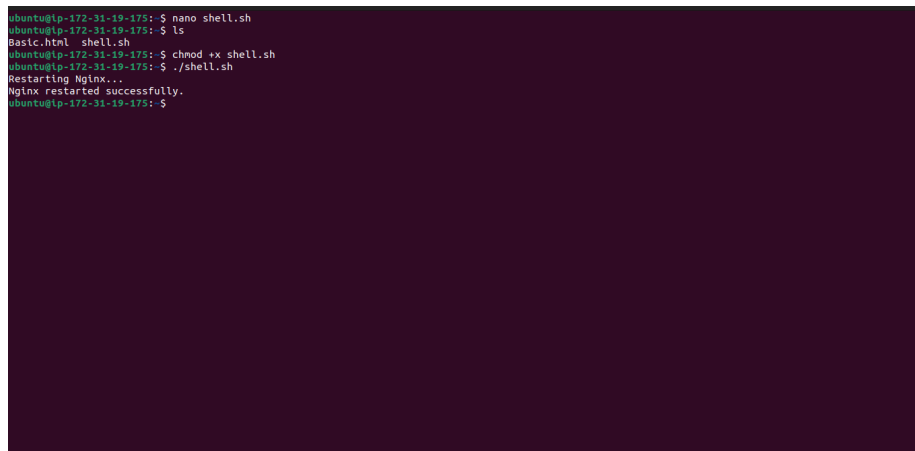


```
GNU nano 8.7.1                                shell.sh *
#!/bin/bash

# Restart Nginx service
echo "Restarting Nginx..."
sudo systemctl restart nginx

if [ $? -eq 0 ]; then
    echo "Nginx restarted successfully."
else
    echo "Failed to restart Nginx."
fi
```

Figure 14: Shell Script



```
ubuntu@ip-172-31-19-175:~$ nano shell.sh
ubuntu@ip-172-31-19-175:~$ ls
Basic.html  shell.sh
ubuntu@ip-172-31-19-175:~$ chmod +x shell.sh
ubuntu@ip-172-31-19-175:~$ ./shell.sh
Restarting Nginx...
Nginx restarted successfully.
ubuntu@ip-172-31-19-175:~$
```

Figure 15: Shell Script Output

Learnings

- Launching and managing AWS EC2 instances.
- Configuring Security Groups.
- Connecting to EC2 using SSH.
- Installing and managing Nginx.
- Deploying a static website.
- Linux command-line operations.
- Git and GitHub workflow.
- Installing and using Docker.
- Writing and executing shell scripts.
- Basic DevOps deployment practices.